1) $1, 10, 19, \ldots, 2026$ /+8

$9, 18, 27, \ldots, 2034$ /:9

$1, 2, 3, \ldots, 226 \Rightarrow 226$

$100 \cdot 9 - 8 = 900 - 8 = 892$

2) $1^2, 2^2, \ldots, 45^2$

$= 45$ m²

$$\begin{array}{r} 45 \\ \cdot\ 45 \\ \hline 225 \\ 180 \\ \hline 2025 \end{array}$$

3) $243 = 3^5$

$000000, \ldots, 100000 = 2^5 = 32$

$20 : 2 = 0$

$10 : 2 = 0$

$5 : 2 = 1$

$2 : 2 = 0$

$1 : 2 = 1$

010100

20	
10	
5	
2	
1	
0	

100100

$2^4 + 2^2$

010100

$\Rightarrow 3^4 + 3^2 = 81 + 9$

$= 90$

4) $_____ = 9^5 \cdot 10$

109999 1

5) $_____ = 900$

$9\ 10\ 10\ 1\ 1$

6) $2^4 \cdot 3^2 \cdot 5 \cdot 7 \Rightarrow \underline{5}\ \underline{3}\ \underline{2}\ \underline{2} = 60$

$A \xrightarrow{6} B \rightarrow C$

$A \xrightarrow{6} B$
$B \xrightarrow{5} C$
$C \xrightarrow{4} B$
$B \xrightarrow{5} A$

$\Rightarrow 20 \cdot 30 = 600$

7) $\underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad} = 5 \cdot 8 \cdot 8 \cdot 7 = 40 \cdot 56 = 2240$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

$8 \quad 8 \quad 7 \quad \underline{1,3,5,7,9}$

5

8) $\underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad} = 2^5 = 32$

$1\ 1 \qquad\qquad 0\ 0\ 0$

b) $\underline{2^9} = 512$

c) 2^9

d) 2^9

e) 2

g) $\underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}$

$\binom{8}{1} + \binom{8}{3} + \binom{8}{5} + \binom{8}{4}$

$= 8 + 56 + 56 + 8$

$= 16 + 112$

$= 128$

$\frac{8 \cdot 7 \cdot 6}{1 \cdot 2 \cdot 3}$

10) $f: \{1,2,3\} \to \{1,2,3,4\}$

a) $4^3 = 64$

b) $x \neq y \Rightarrow f(x) \neq f(y)$

$1 \to 4$
$2 \to 3$
$3 \to 2$
$4 \to 1$

$\Rightarrow 4! = 24$

c) $3^4 = 81$

d) $f(1) < f(2) < f(3)$

123 234 = 4
124
134

11) $8, 9 \neq 0$

$0, 1, 2, 3, 4, 5, 6, 7$

_ _ _ _ _

$\Rightarrow$ 504

Case 1: _ _ _ _ _
(1) (1) (1) (1) (1)
3 4 3 3 2

$= 27 \cdot 8 = 216$

Case 2: _ _ _ _ _
(1) (1) (1) (1) (1)
4 4 3 3 2

$= 288$

12) $\{1, 2, \ldots, 10\}$

a) 3 -> 4, ..., 10 = 2^{7}

$2^{7} = 128$

b) 1 -> 9
2 -> 7
3 -> 5
4 -> 3
5 -> 1

=> 63 · 15 = 945

63 ·
15

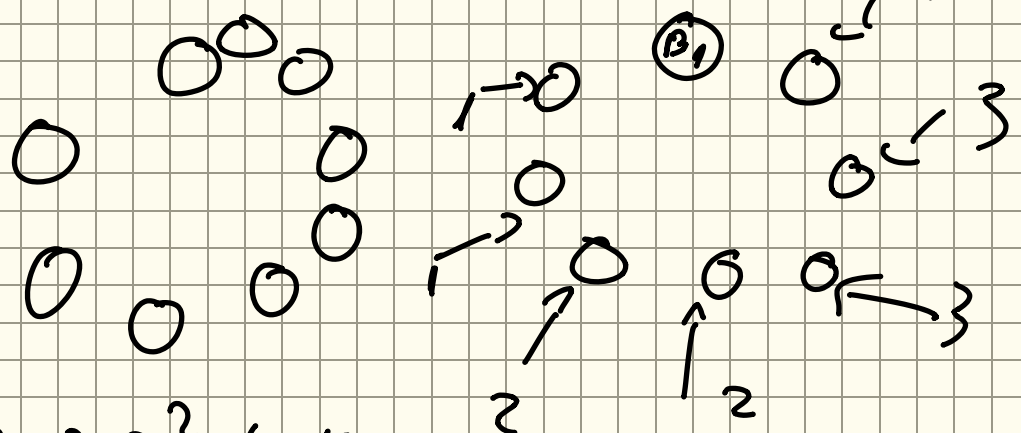
315
63

945

13) (n-1) mecuri

14) $\binom{n+n-1}{n} = \binom{2n-1}{n} =$

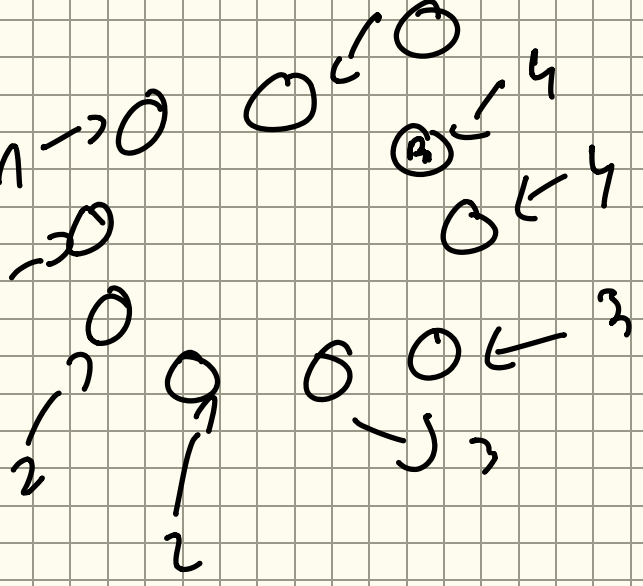
15) a)



$4 \cdot 3 \cdot 3 \cdot 2^{2} = 6 \cdot 12 = 72$

$= \frac{7!}{2} = 3! \cdot 4! = 144$

b)



$4 \cdot 3! \cdot 4! = 576 \cdot 2 = 1152$

16) 1: 1, 2, ..., 9 = 9
2: 9 9 = 81
3: 9 9 8 = 648
⇒ 738

17) ____ Case 1: ___0 = 504
9 8 7

Case 2: ____
8 4 7 4
= 64 · 27
⇒ 2296

18) ____ = 9000
9 10 10 10

____ = 81 · 56
9 9 8 7

56 · 81 / 56 / 448 / 4536

⇒ 4464

26 + 10 = 36

19) ______ ⇒ 36^6
6
______ ⇒ 26^6

⇒ $36^6 - 26^6$

20) $20^{\underline{3}} = 20 \cdot 19 \cdot 18 = 6840$

19 · 18 / 152 / 19 / 342

P: 18
VP: 19 = $19^2 \cdot 18$ =
S: 19

21) $\underline{3}\,\underline{3}\,\underline{3}\,\underline{3}\,\underline{3}\,\underline{3} = 3^6 = 729$ | A, B, C

$\Rightarrow 665$

$\underline{2}\,\underline{2}\,\underline{2}\,\underline{2}\,\underline{2}\,\underline{2} = 2^6 = 64$

$\underline{3}\,\underline{2}\,\underline{2}\,\underline{2}\,\underline{2}\,\underline{2} = 2^5 \cdot 3 = 96$

22) $f: \{1,2,3,4\} \to \{1,2,3,4,5,6\}$

$6^{\underline{4}} = 6 \cdot 5 \cdot 4 \cdot 3 = 30 \cdot 12 = 360$

$f(1) < f(2)$

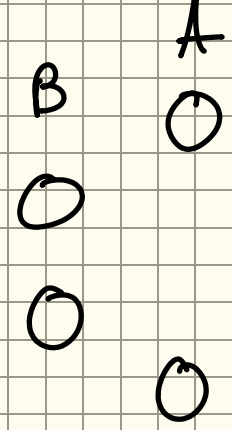
$f(1) = 1 \Rightarrow f(2) = 5$ | $\Rightarrow 5! \cdot 30$

$f(2) = 2 \to f(3) = 4$

$3, 4 = 6 \cdot 6 = 36 \cdot 15$

$= 540$

23)



$\frac{6}{6} = 5! = 120$

$4! + 4! = 48$

$120 - 48 = 72$

$\frac{9^{\underline{6}}}{9} = \frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4}{9} =$

$120 \cdot 56 = 6720$

24) 0, 1, 2, 3, 4, 6, 8

1: $\underset{\downarrow 6}{_}\ \underset{\downarrow 5}{_}\ \underset{\downarrow 4}{_}\ \underset{\downarrow 3}{_}\ \underset{\downarrow 1}{0} = 360$

2: $\underset{\downarrow 5}{_}\ \underset{\downarrow 5}{_}\ \underset{\downarrow 4}{_}\ \underset{\downarrow 3}{_}\ \underset{\downarrow 4}{_} = 400 \cdot 3 = 1200$ $\Rightarrow 1560$

25) $\underset{\downarrow 4}{_}\ \underset{\downarrow 4}{_}\ \underset{\downarrow 4}{_}\ \underset{\downarrow 4}{_} = 16 \cdot 16 = 256$ 1, 2, 3, 4

$\underset{\downarrow 3}{_}\ \underset{\downarrow 3}{_}\ \underset{\downarrow 3}{_}\ \underset{\downarrow 3}{_} = 3^4 = 81$ $\Rightarrow 256 - 81 = 175$

1, 2, ..., 6, ..., ~~9~~

$\underset{5}{_}\ \underset{5}{_}\ \underset{5}{_}\ \underset{5}{_} = 5^4$

$\underset{6}{_}\ \underset{6}{_}\ \underset{6}{_}\ \underset{6}{_} = 6^4 =$

0, ..., 6

$\underset{6}{_}\ \underset{7}{_}\ \underset{7}{_}\ \underset{7}{_} = 7^3 \cdot 6$

$\underset{5}{_}\ \underset{6}{_}\ \underset{6}{_}\ \underset{6}{_} = 6^3 \cdot 5$

26) $M = \{0, \dots, 8\} = 2^9 - 2^5 +$

$\{0,2,4,6,8\}$ $= 512 - 32$

$\{1,3,5,7\}$ $= 480$

$0,2,4 \leq 5$ $\quad \{0,2,4\} = 2^3 - 1 = 7$

$6,8 \geq 5$ $\quad \{6,8\} = 2^2 - 1 = 3$

$\{0,1,2,3,4,5,7\} = 2^7$

2^9 $\quad \{1,3,5,7\}$

$\{1,3,5,6,7,8\}$

$\frac{3}{} \quad \frac{2}{}$

$2^9 - 2^7 - 2^6 + 1 + 2^1 - 1$

$= 512 - 128 - 64 + 16$

$= 512 - 176 = 336$

27) $2^{10} = 1024 - 2 = \frac{1022}{2} = 511$

$\{1\}$ $\quad \{2\}$ $\quad \{3,4,\dots,10\}$

$2^8 = 256$ $\quad \{1,2\}$

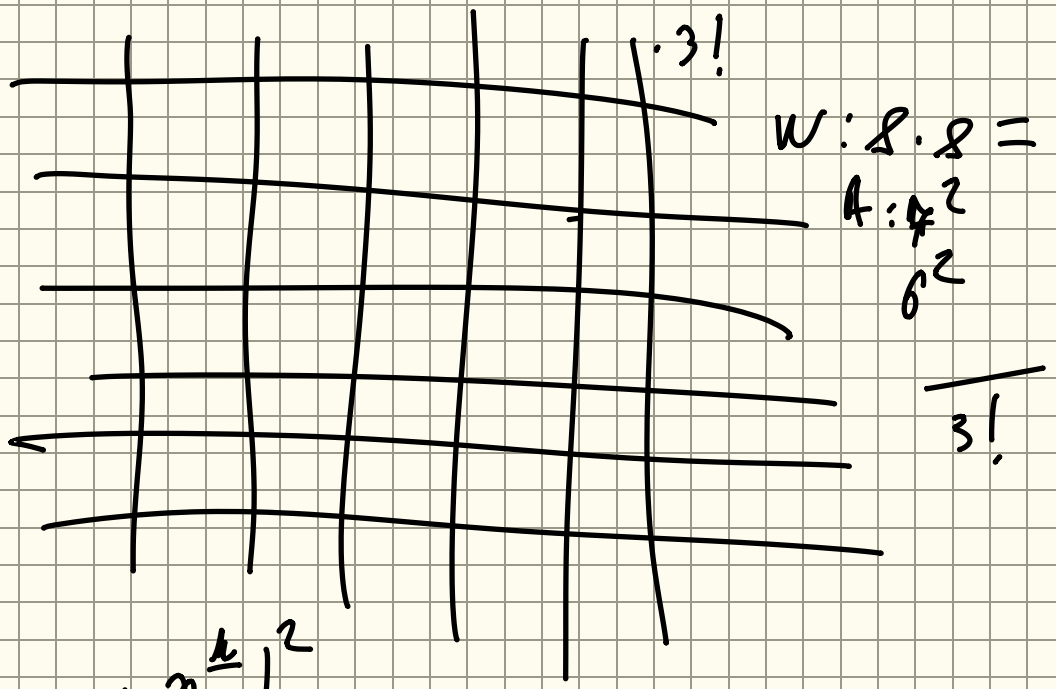
$2^8 = 256$ $\quad 511 - 256 =$

$2^9 = \frac{512}{2} =$ $\quad 8^5 = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 2$

28) $\begin{matrix} 312 \\ 213 \end{matrix}$ $\quad - \quad - \quad = 1440$

$\frac{13440}{2} = 6720$

29) $8!$ $\quad 8^{\underline{3}} = 8 \cdot 7 \cdot 6$



$\cdot 3!$

$W : 8 \cdot 8 =$

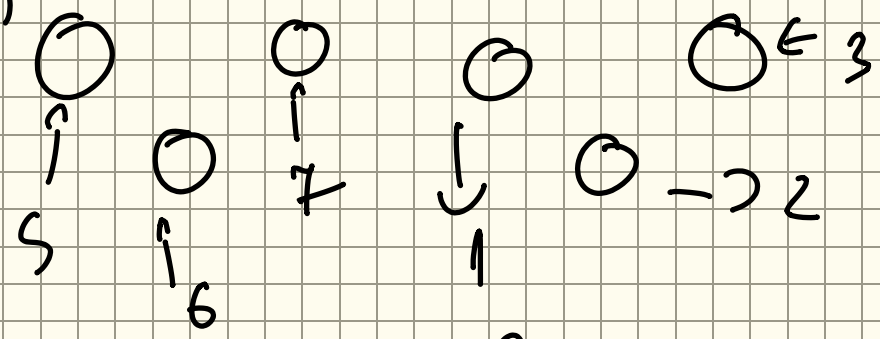
$4 : 4^2$

6^2

$3!$

$\frac{|\eta^{\underline{k}}|^2}{k!}$ $\quad 1, 2, \ldots, 8$ $\quad 1$

30)



$7 \cdot 6 \cdot 5 \cdot 3 \cdot 2 = 36 \cdot 35 =$

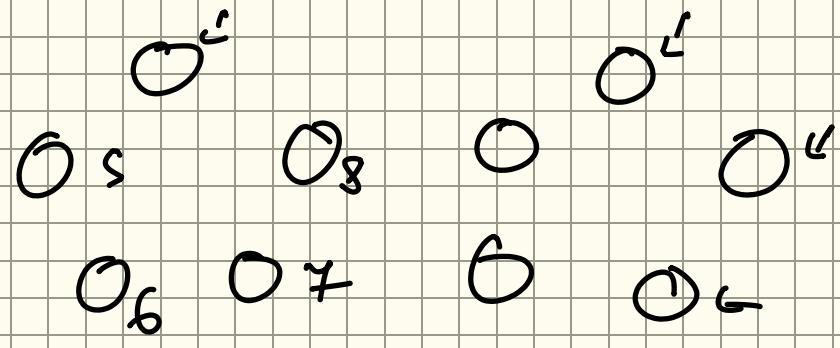
$(n-1)! = 7!$

31)

$x_1 + x_2 + x_3 + x_4 = 11$

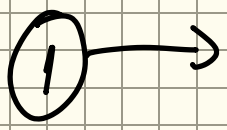
$x_2 = x_2' + 1$

32)



$$\frac{8!}{2} + \frac{4^3}{4} = 3 \cdot 2 = 6$$

$$6 \cdot 4!$$



33) 6S 8P 4 $\binom{14}{4} = \frac{14 \cdot 13 \cdot 12 \cdot 11}{1 \cdot 2 \cdot 3 \cdot 4} = 7 \cdot 13$

$\Rightarrow$

$$\binom{6}{4} + \binom{8}{4} = \frac{6 \cdot 5}{1 \cdot 2} + \frac{8 \cdot 7 \cdot 6 \cdot 5}{1 \cdot 2 \cdot 3 \cdot 4}$$

$$= 15 + 70 = 85$$

$$\binom{6}{3} \cdot \binom{8}{1} = \frac{6 \cdot 5 \cdot 4}{1 \cdot 2 \cdot 3} \cdot 8 = 160$$

$$\binom{6}{4} \cdot \binom{8}{0} = \frac{6 \cdot 5}{1 \cdot 2} = 15$$

34) BANANA $\binom{6}{1,3,2} = \frac{6!}{3!\,2!}$

$$= \frac{6 \cdot 5 \cdot 4}{2} = 60$$

$$\binom{5}{1,1,3} = \frac{5!}{3!} = 5 \cdot 4 = 20$$

35) $x+y+z=15$

$\left(\!\binom{3}{15}\!\right) = \binom{17}{15} = \frac{17 \cdot 16}{1 \cdot 2} = 136$

$x'+y'+z' = 12 \Rightarrow \left(\!\binom{3}{12}\!\right) = \binom{14}{12} = \frac{14 \cdot 13}{2} = 91$

$x+y+z \leq 15 \quad 364 \quad \left(\!\binom{3}{15}\!\right) + \left(\!\binom{3}{14}\!\right) + \ldots + \left(\!\binom{3}{0}\!\right)$

36) $\binom{9}{6} = \frac{9 \cdot 8 \cdot 7}{1 \cdot 2 \cdot 3} = 84$

$\left(\!\binom{9}{6}\!\right)$

$a+b+c+d+e+f=9$

37) $\left(\!\binom{3}{10}\!\right) = \binom{12}{10} = 66$

$\left(\!\binom{3}{4}\!\right) = \binom{6}{4} = 15$

38) $\underline{\quad}\,\underline{\quad}\,\underline{\quad}\,\underline{\quad}$ $\quad 0, \ldots, 9$

$9 \quad 10 \quad 10 \quad 10 \qquad 9000$

$\underline{\quad}\,\underline{\quad}\,\underline{\quad}\,\underline{\quad} = 9 \cdot 9 \cdot 9 \cdot 8$

$8999 \qquad 8172$

$9000 - 5832$

$= 3168$

$\frac{81}{72}$

162

R=0

$\frac{567}{5832}$

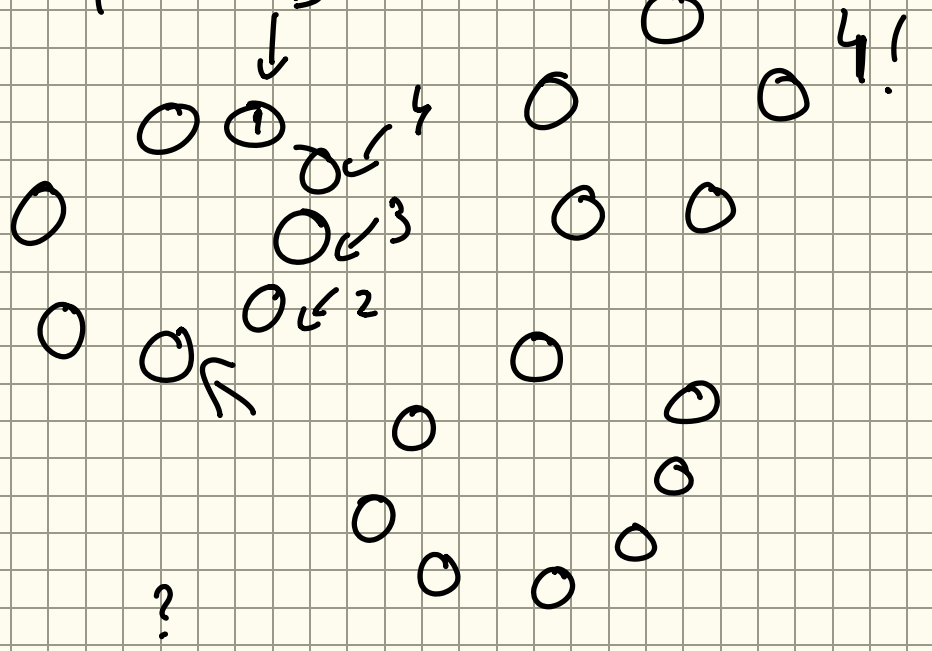
39) $0, 1, 2, 3, 4, 5, 6$ (R: 4)

$\underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad} = 15 \cdot 20 = 300$

$\downarrow\ \downarrow\ \downarrow\ \downarrow$

$5\ \ 5\ \ 4\ \ 3$

40)



41) $\binom{11}{4} - \binom{6}{4} = \frac{11 \cdot 10 \cdot 9 \cdot 8}{1 \cdot 2 \cdot 3 \cdot 4} = 330$

$\binom{6}{4} = \frac{6 \cdot 5}{1 \cdot 2} = 15 \Rightarrow 315$

(R: B)

42) Case 1: BALADA

B: $\binom{5}{3,1,1} \cdot 3 = \frac{5!}{3!} \cdot 3 = 5 \cdot 4 \cdot 3 = 60$

(R: E)

43) $\begin{cases} a+b+c+d+e+f = 8 \end{cases}$

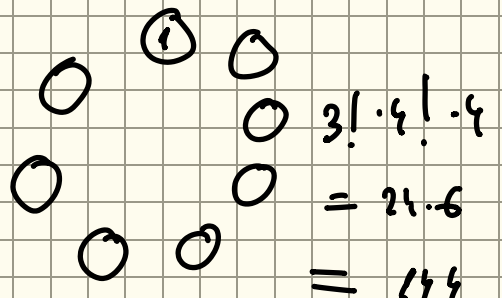
$\binom{6}{3} = \frac{6 \cdot 5 \cdot 4}{1 \cdot 2 \cdot 3} = 20$

$a+b+c = 5$

$\left(\binom{3}{5}\right) \cdot 20$

$\binom{7}{5} \cdot 20 \Rightarrow \frac{7 \cdot 6^3}{1 \cdot 2} = 420$ (R: D)

44)

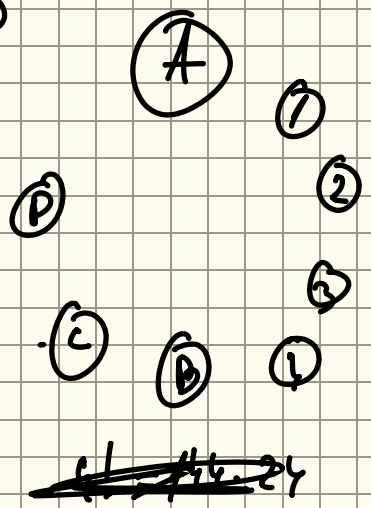


$3! \cdot 4! \cdot 4$

$= 24 \cdot 6$

$= 144$

ABCD
1234



4

~~144 · 24~~

1) ABCDE

$\underline{\quad}\underline{\quad}\underline{\quad} = 125$
$5\,5\,5$

$5 \cdot 4 \cdot 3 = 60$

$5 \cdot 4 \cdot 1 = 20$

$125 - 80 = 45$

$\underline{5}\underline{4}\underline{4} = 80$
$\underline{\quad}\underline{\quad}\underline{5}$

$\Rightarrow 25 \cdot 2 - 5$
$= 45$ (R: C) ✓

2) $\underline{\quad}\underline{\quad}\underline{\quad}$ $1, \ldots, 9$

$\underline{\quad}\underline{\quad}\underline{\quad}$
$\downarrow\downarrow\downarrow$
$7\ 6\ 4$

3) (0 0) (00) (000) $24 \cdot 7 = 168$ (R:B) ✓

$6 \cdot 7 \cdot 4$

$42 \cdot 4 = 8$

$\binom{7}{3} = \frac{7 \cdot 6 \cdot 5}{1 \cdot 2 \cdot 3} = 35$

$\binom{4}{2} = \frac{4 \cdot 3}{2} = 6$

$35 \cdot 6 = 210$ (R:C)

(1,2,3) 4,5,6,7

4) PANORAMA AAAO

(

6) $1 \le p_1 \le p_2 \le p_3 \le p_4 \le p_5 \le 9$

$1 \le 1 \le$

2

3

$p_2 = 3$

$\binom{3}{9} = \binom{11}{9}$

$= \frac{11 \cdot 10}{2} = 55 \cdot 3$

165

$\binom{8}{4} \cdot \binom{4}{1,1,1,1}$

8!

$120 \cdot 6 \cdot 7 \cdot 8$

$720 \cdot 56$

56

720

1120

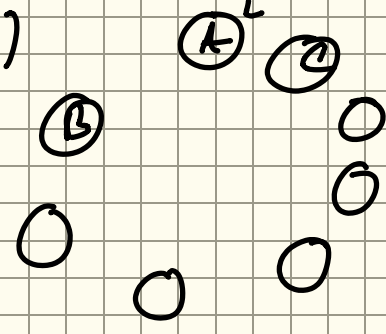
392

40320

PANORAMA

$\binom{8}{4} = 8 \cdot 7 \cdot 6 \cdot 5 = 30 \cdot 56$

$= 1680$ (R:B) ✓

5) 

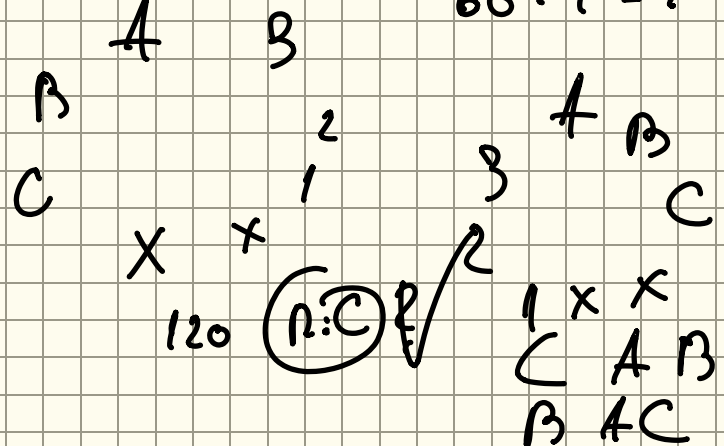
ABC
ACB
CBA

$S^3 = 5 \cdot 4 \cdot 3 = 60 \cdot 2$

n:C ✓

ABC $S^3 = 5 \cdot 4 \cdot 3 = 60 + 60$

$60 \cdot 4 = 240$



120 n:C

6) $1 \leq p_1 \leq 3 \leq p_3 \leq p_4 \leq p_5 \leq 9 \quad /-2$

1
2
3

$1 \leq p_3 \leq p_4 \leq p_5 \leq 4$

$\left(\left(\begin{smallmatrix} 3 \\ 2 \end{smallmatrix}\right)\right) = \binom{9}{2} = \frac{9 \cdot 8}{1 \cdot 2} = 36$

$0 \leq p_1 \leq 2 \leq p_3 \leq p_4 \leq p_5 \leq 8$

0
1
2

$36 \cdot 3 = 108$

$2 \leq p_3 \leq p_4 \leq p_5 \leq 8 \quad /-2$

$0 \leq p_3 \leq p_4 \leq p_5 \leq 6$

$\left(\left(\begin{smallmatrix} 3 \\ 6 \end{smallmatrix}\right)\right) = \binom{8}{6} = 32$

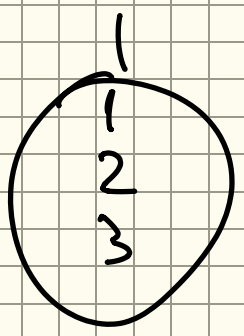
$32 \cdot 3 = 96$

$1 \leq p_1 \leq 3 \leq \ldots \leq 9 \quad /-1$

$0 \leq p_1 \leq 2 \leq \ldots \leq 8$

$1 \le p_1 \le \dots \le 9$

$1 \le p_1 \le 3$.



$3 \le p_3 \le p_4 \le p_5 \le 9$ /-3

$\left(\!\binom{3}{6}\!\right) = \binom{8}{6}$

$= \frac{8 \cdot 7}{2} = 28$

$3 \le p_3 \; p_4 \; p_5 \le 9$ /-2

$\binom{7}{3} = \frac{9 \cdot 8 \cdot 7}{3 \cdot 2} = 36 \cdot 3$

$= 108$

$a+b+c=6$
$a+b+c+d=6$

$\binom{9}{6} = \frac{9 \cdot 8 \cdot 7}{3 \cdot 2}$

84

$3 \le \dots \le 9$

$\binom{3}{9} = 1$

$\binom{7}{3} \; \binom{9}{2} \; \frac{9 \cdot 8}{2}$ 252